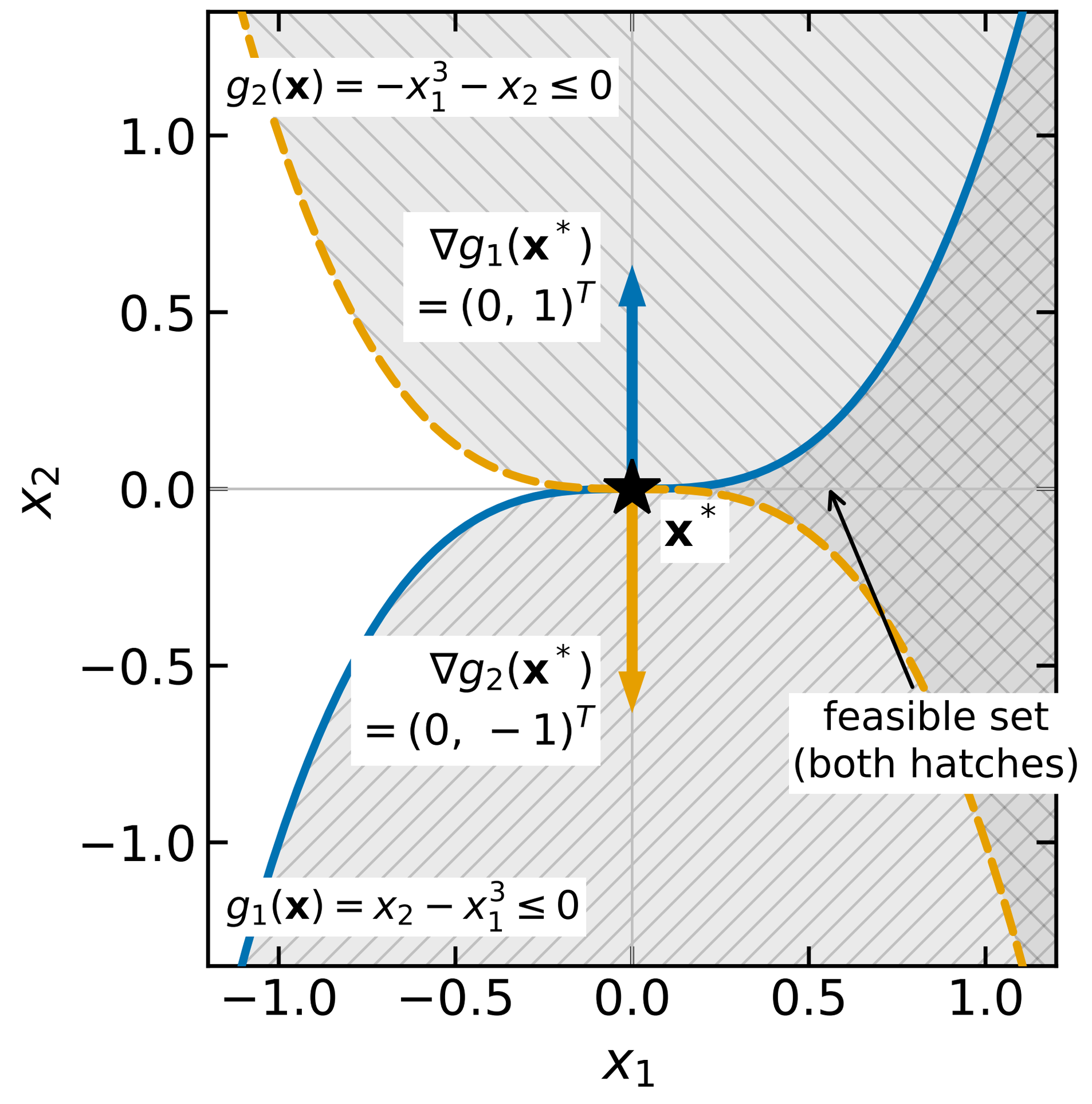
$g_2(\mathbf{x}) = -x_1^3 - x_2 \leq 0$
$\nabla g_1(\mathbf{x}^*) = (0,\ 1)^T$
$\mathbf{x}^*$
$\nabla g_2(\mathbf{x}^*) = (0,\ -1)^T$
feasible set (both hatches)
$g_1(\mathbf{x}) = x_2 - x_1^3 \leq 0$
1.0
0.5
0.0
−0.5
−1.0
x_2
−1.0
−0.5
0.0
0.5
1.0
x_1